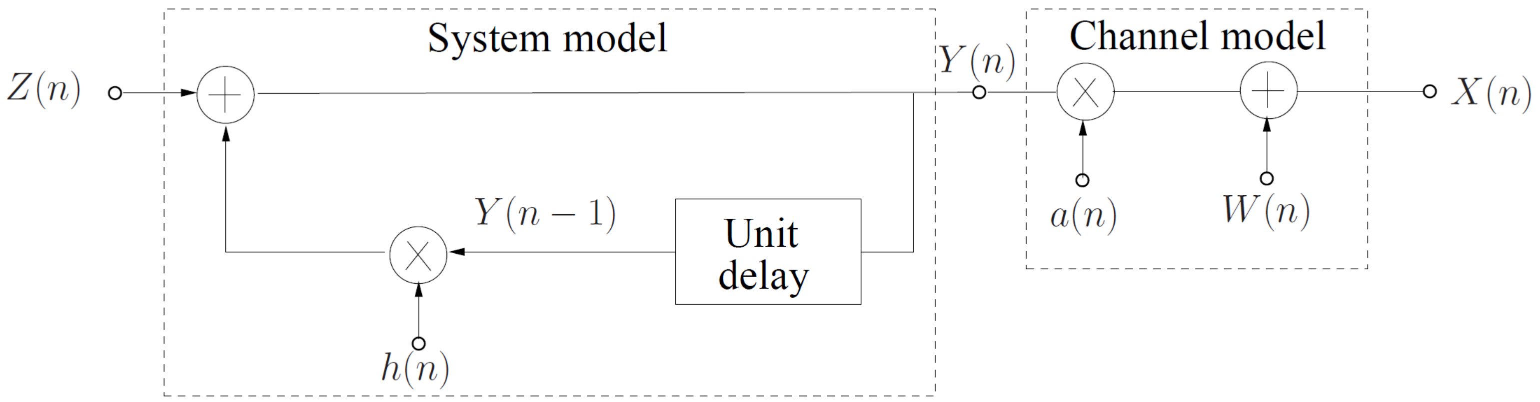
# The Kalman Filter

**Goal:** to find LMMSE estimates of the samples of an unobservable process $Y(n)$ from the samples of a related, observable process $X(n)$, for $n = 1,2, \ldots$

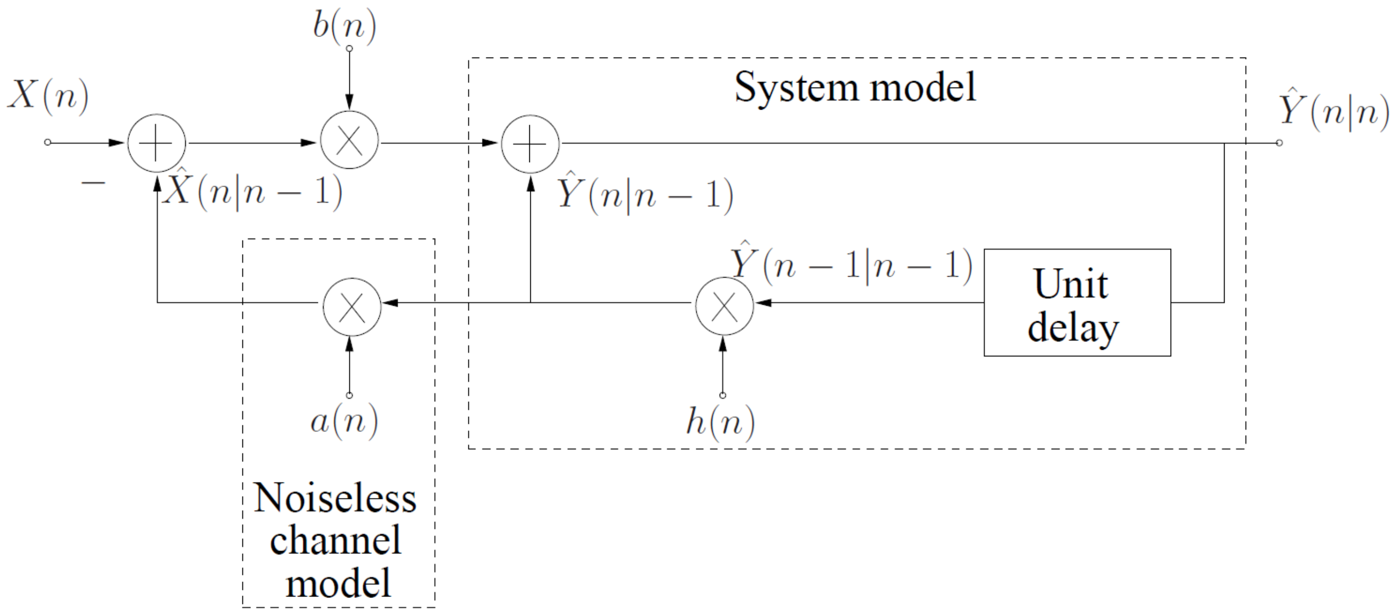
**Recursive implementation:** at time $n + 1$, the LMMSE estimate of $Y(n + 1)$ is obtained as a linear function of the LMMSE estimate of $Y(n)$ and the new observation $X(n + 1)$.

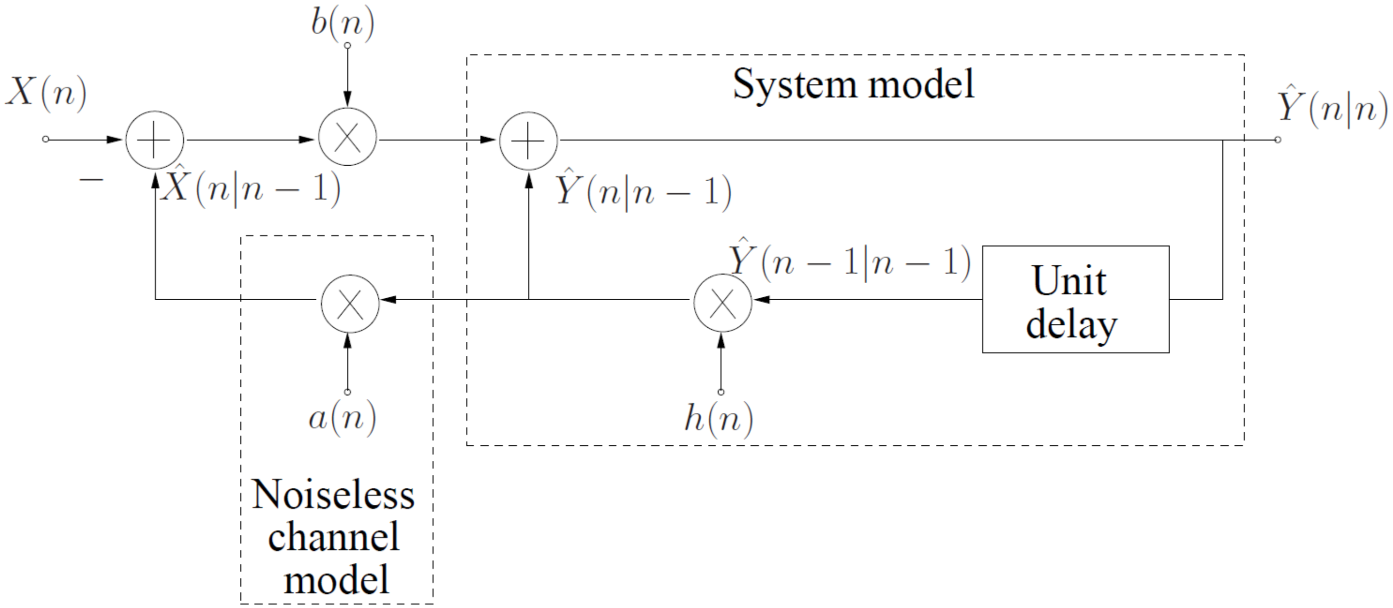
This recursive implementation is only possible when the processes $Y(n)$ and $X(n)$ are defined appropriately.

System and Channel Model for the Kalman Filter:




Block Diagram of the Kalman Filter:





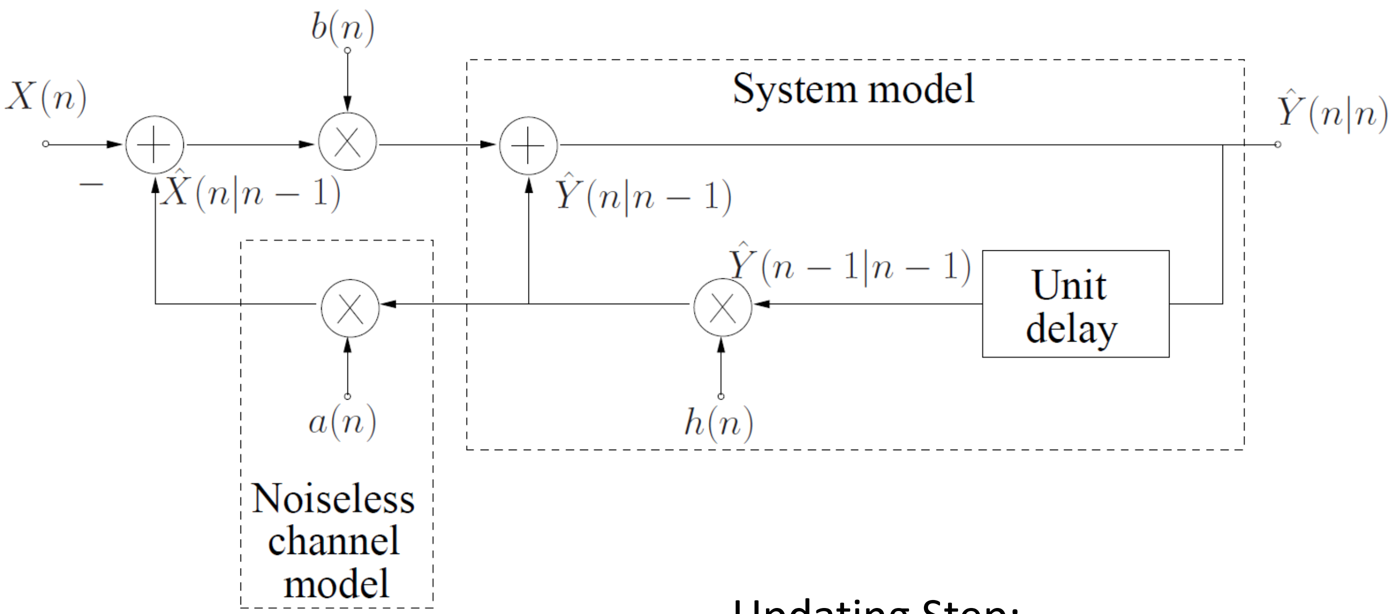

$\hat{Y}(n-1|n-1)$: LMMSE Estimate of $Y(n-1)$ given the observations $X(1), X(2), \ldots, X(n-1)$.

$\hat{Y}(n|n-1)$: LMMSE Estimate of $Y(n)$ given the observations $X(1), X(2), \ldots, X(n-1)$.

$\hat{X}(n|n-1)$: LMMSE Estimate of $X(n)$ given the observations $X(1), X(2), \ldots, X(n-1)$.

$\hat{Y}(n|n)$: LMMSE Estimate of $Y(n)$ given the observations $X(1), X(2), \ldots, X(n)$.

<u>Recursive structure of the Kalman filter:</u> $\hat{Y}(n|n) = \mathcal{LF}\big(\hat{Y}(n-1|n-1), X(n)\big)$




<u>Updating Step:</u>

$$\hat{Y}(n|n) = \hat{Y}(n|n-1) + b(n)\big(X(n) - \hat{X}(n|n-1)\big)$$

with the Kalman gain:

$$b(n) = \frac{a(n)R(n|n-1)}{a(n)^2R(n|n-1)+\sigma_W^2(n)}.$$

In addition, the MSE of $\hat{Y}(n|n)$ is:

$$R(n|n) = \big(1 - b(n)a(n)\big)R(n|n-1)$$

<u>Prediction Step:</u>

$$\hat{Y}(n|n-1) = h(n)\hat{Y}(n-1|n-1),$$

$$\hat{X}(n|n-1) = a(n)\hat{Y}(n|n-1).$$

In addition, the MSE of $\hat{Y}(n|n-1)$ is:

$$R(n|n-1) = h(n)^2R(n-1|n-1) + \sigma_Z^2(n)$$

<u>Initialization:</u> the Kalman filter is initialized by setting $\hat{Y}(0|0) = \mu_{Y(0)}$ and $R(0|0) = \sigma_{Y(0)}^2$